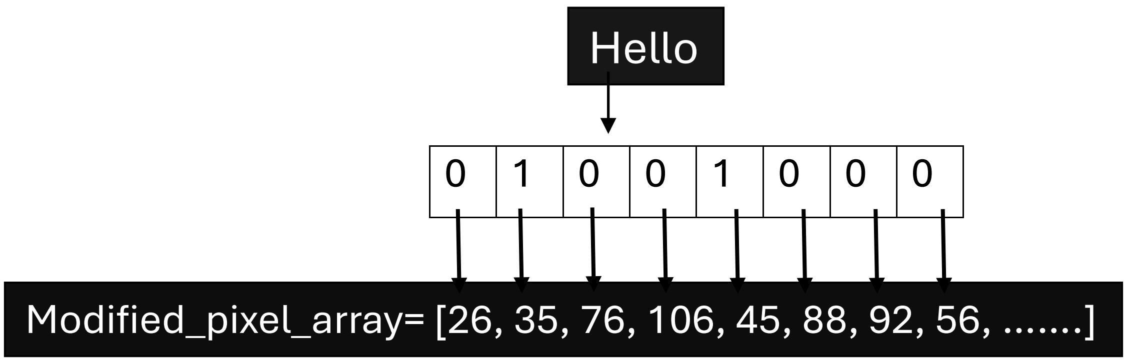
StegoScope

Project Report

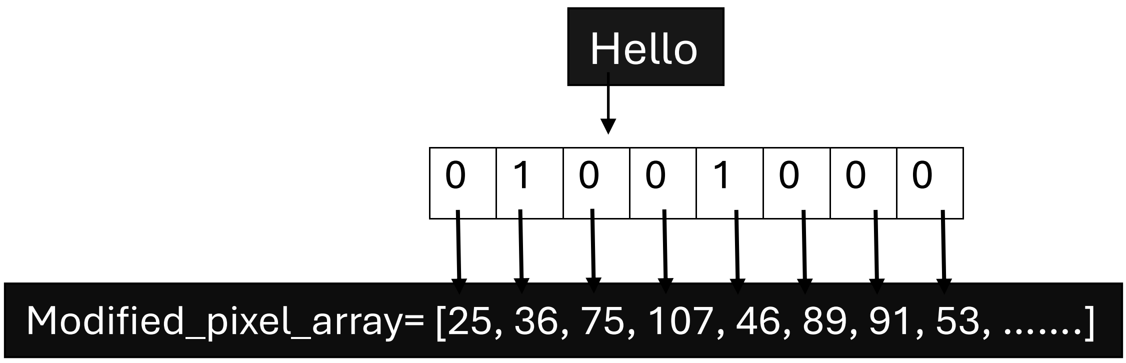
Introduction: StegoScope is a project which can detect the type of file provided by the user and redirects it to the required steganography methods of the file. The project supports image, audio, and video files and embeds/extracts/analyzes the file according to the choice of the user.

Components: The project comprises of the files: ‘detect’, ‘imagesteg’, ‘imagestegcustom’, ‘audiosteg’, ‘videosteg’ and ‘steganalysis’.

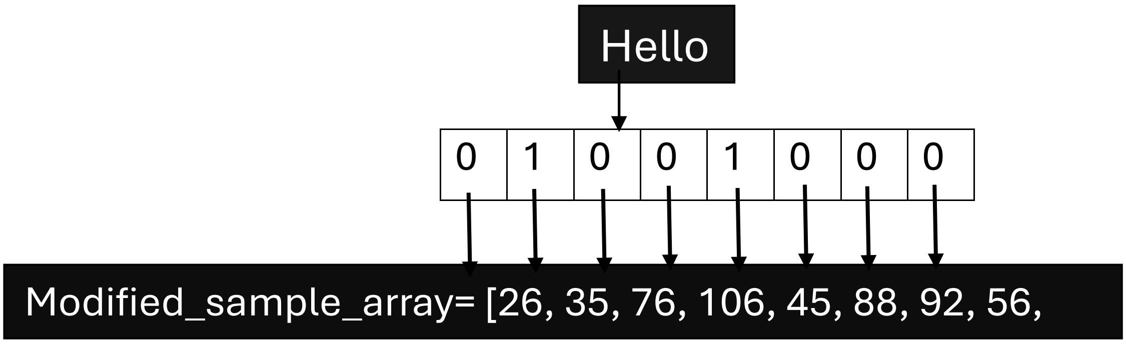
1. **“DETECT”:** The program takes the path of the file, choice(embed/extract/analyze), and the type of output(verbose/quiet). It detects the type of the input file by comparing with the file signatures/magic numbers of file and redirects it to the required steganography programs based on the type of file.
2. **“IMAGESTEG”:** The program embeds the data by changing the LSB (Least Significant Bit) of the pixels of the image by dealing with 3 pixels at a time and the unused value will act as a flag for end of data and extracts the data using the reverse method. And makes a new image using the modified pixel array.
   For data, say, “Hello”, it converts the characters of data into binary string and changes the LSB of the pixels according to the binary string.



3. **“IMAGESTEGCUSTOM”:** The program embeds the data by changing the LSB (Least Significant Bit) of the pixels of the image by the opposite way by dealing with 3 pixels at a time and the unused value will act as a flag for end of data by dealing with 3 pixels at a time and the unused value will act as a flag for end of data by dealing with 3 pixels at a time and the unused value will act as a flag for end of data. And makes a new image using the modified pixel array.



4. **“AUDIOSTEG”:** The program converts the frames of the audio using wave module and converts them into samples by comparing the sample width of the file. It embeds the data by changing the LSB (Least Significant Bit) of the samples of the audio. It also embeds an EOF flag (“11111110”) with the data. And it makes a new audio file using the modified sample array. It uses the reverse code to extract the data from the embedded audio file.



5. **“VIDEOSTEG”:** The program converts the video into its frames using ffmpeg and gets the information about only the key frames (i.e., I-frames) and embeds the data only into these keyframes using the same logic in the image steganography and makes a new video using these modified frames. It extracts the data from the video using the reverse code.
6. **“STEGANALYSIS”:** This folder comprises of various analysis methods which will be implemented to the file according to the file type. This provides the feature of plugins where we can add new steganalysis methods without changing the core program. It uses LSB Histogram test, Chi Square test and RS Analysis for images, LSB Histogram test for audios, and key frame LSB test for videos.